

ECE 2210: CI: Introduction to Programming in Python

HW2 – Arithmetic Operators and Functions

Complete the **four** functions given in the Python scaffold `hw2.py`. The headers of all four functions are given. The requirements of completing these functions are given in the triple-quoted comments above each function. The following information is needed to complete the coding. Note that you are **NOT** allowed to use scientific notation in your code.

- a) Function I & II: `cal_secs_ball_falling1()` and `cal_secs_ball_falling2()`

The standard kinematic formula is $s = \frac{1}{2}gt^2$, where $g = 9.81\text{ms}^{-2}$ is the acceleration due to gravity.

- b) Function III: `cal_altitude()`

The altitude h above the earth's surface that the satellite must have is given by

$$h = \left(\frac{GMT^2}{4\pi^2} \right)^{1/3} - R$$

where $G = 6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ is Newton's gravitational constant, $M = 5.97 \times 10^{24} \text{ kg}$ is the mass of the Earth, and $R = 6371 \text{ km}$ is its radius. The unit of h is m (**not** km).

- c) Function IV: `draw_grid()`

The following graph shows the grid printed by function “`draw_grid`”. You are **required** to use string multiplication and string concatenation for this one. You will receive 0 for this problem without proper use of the related string operations.

```
+ - - - - + - - - - +
|           |           |
|           |           |
|           |           |
+ - - - - + - - - - +
|           |           |
|           |           |
|           |           |
+ - - - - + - - - - +
```

Hint: to print more than one value on a line, you can print a comma-separated sequence of values:

```
print('+', '-')
```

By default, `print` advances to the next line, but you can override that behavior and put a space at the end, like this:

```
print('+', end = ' ')
print('-')
```

The output of these statements is ‘+ -’. A `print` statement with no argument ends the current line and goes to the next line.

Code testing:

The correct output is given in file “`output`”. You can use any text editor (e.g., Sublime) to open it to see what the correct output looks like. To test your code, run it in the Linux terminal

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using the following command, which generates a text file called “**myoutput**” that contains the output of your program.

```
$ python3 hw2.py > myoutput
```

Then, run the following command in Linux terminal to compare the output of your program and the given output.

```
$ diff output myoutput
```

If there is no output of running **diff**, it means the **output** and **myoutput** are identical. Note that TA will use the same command to test your program.

Submission: For the submission of this assignment, you need to upload the completed **hw2.py** to Canvas.

Grading: The output(s) of each function is worth 25 points.

Warning: The TA will compare your code with and your classmates’ codes using MOSS (<https://theory.stanford.edu/~aiken/moss/>). A similarity of 40% or higher is considered as plagiarism. The best way to avoid a high similarity is to write the code all by yourself. Discussion is always encouraged but never look at others’ code.